Review plan · for the lecturer

Teaching machine learning at university: unit check

AI · University · Unit check

Time
40 min

Materials
Projector or board, and the lecture slides.

Learning goals

- Optimisation
- Backpropagation
- Attention
- Evaluation and assessment

Activity: Peer-instruction review

1. Vote (10 min): Show each question; students answer alone and commit to an answer.
2. Discuss (15 min): Where answers split, pairs argue for two minutes and vote again.
3. Explain (10 min): Work through the questions most students missed, showing the reasoning step by step.
4. Plan (5 min): Note which misconception to revisit next week and how.

Discussion

- Which misconception will you target first?
- Which lab will you add?
- How will you change assessment?

Exit ticket

- One change to my ML course this term, and the evidence I will collect.

Try it in your next lecture
Bring one revised ML assignment and its rubric to your next department meeting.

Quiz sheet · for students

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Name _____ Group _____ Date _____

1 **Loss θ^2, $\theta = 5$, $\eta = 0.1$. After one step, θ =**

◯ 3 ◯ 0 ◯ 4 ◯ 4.9

2 **The negative gradient points directly at the minimum.**

◯ True ◯ False

3 **Apply the chain rule.**

$\partial L/\partial a = 4$ and $\partial a/\partial w = 0.5$, so $\partial L/\partial w =$ _____ .

Word bank: 4.5 · 8 · 2

4 **Which reduce vanishing gradients? Choose all that apply.**

☐ ReLU ☐ Residual connections ☐ Deeper sigmoid stacks

5 **d = 256. Raw attention scores are divided by**

◯ 256 ◯ 16 ◯ 8

6 **Match each transformer part to its job.**

Positional encoding	• • • •	Different relations
Causal mask	• • • •	Word order
Multiple heads	• • • •	No peeking ahead

7 **Put an honest evaluation in order.**

☐ Tune on validation folds

☐ Score once on the test set

☐ Split off a test set

☐ Pick the final model

8 **A model with 99% accuracy can have 0% recall.**

◯ True ◯ False

9 **Which best assesses a student's own understanding of their ML project?**

◯ Code length ◯ An oral defence of their notebook ◯ The final accuracy alone

10 **What belongs in a model card? Choose all that apply.**

☐ Intended use ☐ Metrics by subgroup ☐ The team's leaderboard rank ☐ Known limitations

Answers · for the lecturer

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1 **4**

$\theta - \eta \cdot 2\theta = 5 - 1 = 4$.

2 **False**

It points in the steepest local descent direction only.

3 **$\partial L/\partial a = 4$ and $\partial a/\partial w = 0.5$, so $\partial L/\partial w = \underline{2}$.**

$4 \times 0.5 = 2$.

4 **ReLU; Residual connections**

Both keep local derivatives near 1 along some path.

5 **16**

$\sqrt{256} = 16$.

6 **Positional encoding — Word order**
Causal mask — No peeking ahead
Multiple heads — Different relations

Attention alone ignores word order, so positional encoding adds it; the causal mask stops a token looking at later tokens; and several heads capture different relations at once.

7 **Top to bottom: 2, 4, 1, 3**

The test set is touched once, at the end.

8 **True**

With 1% positives, predicting all negative gives exactly that.

9 **An oral defence of their notebook**

A defence probes reasoning a tool cannot supply.

10 **Intended use; Metrics by subgroup; Known limitations**

Model cards document use, data, disaggregated metrics and caveats.